

BHTTP/1 — annotated hexdump of one complete exchange

Captured from ./bcurl -v localhost:9000/index.html against ./bserve ./www 9000. Raw capture: docs/capture.txt

Here is everything that crossed the TCP connection. The client sent one frame of 61 octets. The server sent back two frames, 89 + 47 = 136 octets, and the connection then stayed open. Offsets count octets from the start of each direction's byte stream. Each frame starts with the same 8-octet header: Length (2 octets), Type (1), Flags (1), and Request ID (4). After the header come exactly Length octets of payload. Inside a HEADERS payload, every field is a prefix octet, a 2-octet value length, and then the value.

Colour key: frame header field prefix (static index) value length value / body octets

Client → server: frame 1, HEADERS (the request) — 61 octets

0000	00 35 01 01 00 00 00 01	01 00 03 47 45 54 02 00	.5.....GET..
0010	0b 2f 69 6e 64 65 78 2e	68 74 6d 6c 04 00 0e 6c	./index.html...
0020	6f 63 61 6c 68 6f 73 74	3a 39 30 30 30 05 00 07	localhost:9000...
0030	62 63 75 72 6c 2f 31 06	00 03 2a 2f 2a	bcurl/1...*/
Offset	Octets	Field	Meaning
0000–0001	00 35	Length	0x0035 = 53: payload octets after this 8-octet header
0002	01	Type	0x01 = HEADERS
0003	01	Flags	0x01 = END: no request body follows
0004–0007	00 00 00 01	Request ID	1: first request on this connection
0008	01	prefix	0x01: static index 1 = :method (name costs 1 octet)
0009–000a	00 03	value length	0x0003 = 3
000b–000d	47 45 54	value	"GET"
000e	02	prefix	0x02: static index 2 = :path (name costs 1 octet)
000f–0010	00 0b	value length	0x000b = 11
0011–001b	2f 69 6e 64 65 78 2e 68 74 6d 6c	value	"/index.html"
001c	04	prefix	0x04: static index 4 = host (name costs 1 octet)
001d–001e	00 0e	value length	0x000e = 14
001f–002c	6c 6f 63 61 6c 68 6f 73 74 3a 39 30 ... (14 octets)	value	"localhost:9000"
002d	05	prefix	0x05: static index 5 = user-agent (name costs 1 octet)
002e–002f	00 07	value length	0x0007 = 7
0030–0036	62 63 75 72 6c 2f 31	value	"bcurl/1"
0037	06	prefix	0x06: static index 6 = accept (name costs 1 octet)
0038–0039	00 03	value length	0x0003 = 3
003a–003c	2a 2f 2a	value	"*//*"

Server → client: frame 1, HEADERS (the response head) — 89 octets

0000	00 51 01 00 00 00 00 01	03 00 03 32 30 30 09 00	.Q.....200..
0010	08 62 73 65 72 76 65 2f	31 0a 00 1d 54 68 75 2c	.bserve/1...Thu,
0020	20 30 38 20 4f 63 74 20	32 30 32 36 20 30 39 3a	08 Oct 2026 09:
0030	33 34 3a 31 36 20 47 4d	54 07 00 18 74 65 78 74	34:16 GMT...text
0040	2f 68 74 6d 6c 3b 20 63	68 61 72 73 65 74 3d 75	/html; charset=u
0050	74 66 2d 38 08 00 02 33	39	tf-8...39
Offset	Octets	Field	Meaning
0000–0001	00 51	Length	0x0051 = 81: payload octets after this 8-octet header
0002	01	Type	0x01 = HEADERS
0003	00	Flags	0x00: END clear, so DATA frames follow
0004–0007	00 00 00 01	Request ID	1: first request on this connection
0008	03	prefix	0x03: static index 3 = :status (name costs 1 octet)
0009–000a	00 03	value length	0x0003 = 3

000b–000d	32 30 30	value	"200" — three ASCII digits
000e	09	prefix	0x09: static index 9 = server (name costs 1 octet)
000f–0010	00 08	value length	0x0008 = 8
0011–0018	62 73 65 72 76 65 2f 31	value	"bserve/1"
0019	0a	prefix	0x0a: static index 10 = date (name costs 1 octet)
001a–001b	00 1d	value length	0x001d = 29
001c–0038	54 68 75 2c 20 30 38 20 4f 63 74 20 ... (29 octets)	value	"Thu, 08 Oct 2026 09:34:16 GMT" — RFC 1123 time, 29 octets
0039	07	prefix	0x07: static index 7 = content-type (name costs 1 octet)
003a–003b	00 18	value length	0x0018 = 24
003c–0053	74 65 78 74 2f 68 74 6d 6c 3b 20 63 ... (24 octets)	value	"text/html; charset=utf-8" — chosen from the .html extension
0054	08	prefix	0x08: static index 8 = content-length (name costs 1 octet)
0055–0056	00 02	value length	0x0002 = 2
0057–0058	33 39	value	"39" — advisory; END is what ends the body

Server → client: frame 2, DATA (the body) — 47 octets

0059	00 27 00 01 00 00 00 01	3c 21 64 6f 63 74 79 70	.'.....<!doctype
0069	65 20 68 74 6d 6c 3e 0a	3c 68 31 3e 68 65 6c 6c	e html>.<h1>hell
0079	6f 2c 20 62 69 6e 61 72	79 3c 2f 68 31 3e 0a	o, binary</h1>.

Offset	Octets	Field	Meaning
0059–005a	00 27	Length	0x0027 = 39: payload octets after this 8-octet header
005b	00	Type	0x00 = DATA
005c	01	Flags	0x01 = END: last frame of this response
005d–0060	00 00 00 01	Request ID	1: first request on this connection
0061–0087	3c 21 64 6f 63 74 79 70 65 20 68 74 ... (39 octets)	body	the 39 octets of www/index.html, sent unchanged: '<!doctype html>\n hello, binary \n'

What the octets add up to

	BHTTP/1	Same message as HTTP/1.1 text	Notes
request	61	84	8 frame header + 5 × 3 field overhead + 38 value octets
response head	89	134	8 frame header + 5 × 3 field overhead + 66 value octets
response body	8 + 39	39	one DATA frame; END marks the end, so no chunked encoding is needed

Every header name in this exchange was in the static table, so each name cost exactly one octet. A header outside the table, such as x-trace: 7, would be sent literally:

00 07 "x-trace" 00 01 "7"

That is a literal marker, the name's length, the name, the value's length and the value. If -g had been given, an extra frame such as 00 03 ba ff 00 00 00 01 "v2?" would sit in front of the request. Its type 0xBA is unassigned, so the server reads its 3 payload octets, discards them, and answers exactly as shown above.